Yahan Pu

E-Mail: yahanpu.astro@gmail.com **Website:** yahanpu.github.io **TEL:** +86 18008210240

EDUCATION

University of Science and Technology Beijing **Beijing, China**
Bachelor of Science in Applied Physics, GPA: 3.55/4.00 Sept 2022 – Jun 2026
(Estimated German equivalent: 1.4, Modified Bavarian Formula)

- Mathematical Analysis, Methods of Mathematical Physics, Ordinary Differential Equations, Probability and Mathematical Statistics, Quantum Mechanics, Electrodynamics, Atomic Physics, Thermodynamics and Statistical Physics.

University of California, Berkeley **Berkeley, CA**
Visiting Student, GPA: 4.00/4.00 Sep 2025 – Dec 2025

- Astronomy Data Science Laboratory, Stellar Physics, Planetary Astrophysics.

RESEARCH EXPERIENCE

Research on Galaxy Kinematics and Morphology in IllustrisTNG Sep 2024 – Present
National Astronomical Observatories, Chinese Academy of Sciences Beijing, China
Advisor: Prof. Lan Wang

Photometric and Kinematic Galaxy Morphology in IllustrisTNG

- Comparing photometric B/T estimates derived from mock galaxy images across different decomposition models and viewing directions.
- Investigating systematic biases associated with stellar mass, inclination, and profile assumptions to inform photometric model selection.

Galaxy Merger Orbits and Baryonic Effects in IllustrisTNG

- Constructed one-to-one matched merger samples between hydrodynamical and dark-matter-only simulations.
- Analyzed merger timescales and orbital evolution across different galaxy masses, merger mass ratios, and numerical resolutions.
- Found shorter merger timescales and relatively more spiral-in orbital evolution in hydrodynamical simulations than in dark-matter-only counterparts.

Research on Viscoelastic Thermal Convection in Polymer Solutions Oct 2023 – Sep 2024
School of Mathematics and Physics, University of Science and Technology Beijing Beijing, China
Advisor: Prof. Xinhui Si

- Modified C++/OpenFOAM solvers for viscoelastic thermal convection, including governing equations, boundary conditions, and numerical components; performed mesh and time-step convergence tests.
- Analyzed simulation results to quantify heat transport and flow behavior.
- Contributed to debugging, numerical validation, and figure preparation for a companion study on viscoelastic convection around an elliptical cylinder.

PUBLICATIONS

- **Pu, Y.**, Wang, L., Zeng, G., & Xie, L. (2026). Comparing Galaxy Merger Orbits in Hydrodynamical Simulation and in Dark Matter only Simulation. *Research in Astronomy and Astrophysics*, 26(11), 115009.
- **Pu, Y.**, Guo, B., & Si, X. (2025). Natural convection of dilute polymer solution in a differentially heated square cavity with different thermal boundary conditions. Physics of Fluids, 37(9).
- Guo, Y., Guo, B., **Pu, Y.**, & Si, X. (2025). Numerical investigation of viscoelastic thermal convection in entangled polymer solutions: Elliptical cylinder effects within a square enclosure. Physics of Fluids, 37(12).

EXTRACURRICULAR ACTIVITIES

Peer Physics Tutor, School of Mathematics and Physics Sep 2023 – Sep 2024

- Led monthly physics review sessions for junior undergraduate students and provided one-on-one tutoring.

President, Student Physics Association Sep 2024 – Sep 2025

- Organized undergraduate physics seminars with invited faculty speakers.

SKILLS

Technical Skills: Python (PyTorch, Astropy), C++, MATLAB; MCMC, statistical analysis; IllustrisTNG, OpenFOAM; Linux, LaTeX.
Language Skills: English (TOEFL iBT: 102: R 29, L 26, S 23, W 24), German (Beginner), Chinese (Native).